

Wednesday Warm Up: Unit 0: Electrostatics I and II

Prof. Jordan C. Hanson

January 29, 2025

1 Memory Bank

- $1e = 1.602 \times 10^{-19} \text{ C}$... Fundamental unit of charge.
- $\vec{E} = \frac{kQ}{r^2} \hat{r}$... Coulomb field of a charge Q .
- $k = 8.988 \times 10^9 \text{ N C}^{-2} \text{ m}^2$... Constant of proportionality for Coulomb field.
- $\vec{F} = q\vec{E}$... The force on a charge q by a field $\vec{E}$.

2 Electric Fields

1. Consider Fig. 1 below. A ring of charge with radius R is situated in the xy -plane. The charge is positive, and it is distributed evenly across the ring. We write $\Delta q = \lambda R \Delta \theta$, to mean that there are λ Coulombs per meter along an angle $\Delta \theta$. If $\Delta \theta$ were to extend to 2π (all the way around the circle), then the total charge is $Q = \lambda(2\pi R)$. (a) By *symmetry*, where should the electric field be zero? (b) Where would the electric field be infinite?

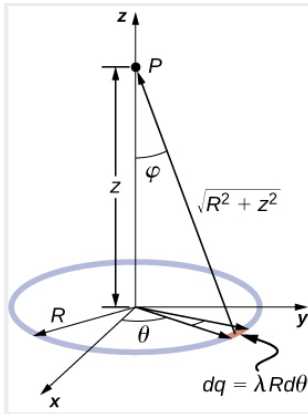


Figure 1: A ring of charge situated in the xy -plane.

2. Consider again Fig. 1. As $z \rightarrow \infty$, what happens to the field?
 - A: The field-strength increases.
 - B: The field-strength remains constant.
 - C: The field-strength decreases.
 - D: The field-strength is exactly zero.

3. Suppose the actual function for the E-field $\vec{E}(z)$ is

$$\vec{E}(z) = \frac{kQz}{(z^2 + R^2)^{3/2}} \hat{z} \quad (1)$$

To see what happens when z is much larger than R , try setting $R = 0$. What is the result in Eq. 1 if $R = 0$?

4. Do you recognize the expression? What does the ring resemble if $R \approx 0$?

5. Suppose a test-charge $q = 1 \text{ nC}$ is placed at $z = 20 \text{ cm}$ above a ring of radius $R = 2 \text{ cm}$. (a) What is the value $\vec{E}$, in N C^{-1} ? (b) What is the force on q ? (c) How many units of charge constitute q ?